

Agenda

• Decision-Making

• Quantitative Vs Qualitative

• Learning from Data

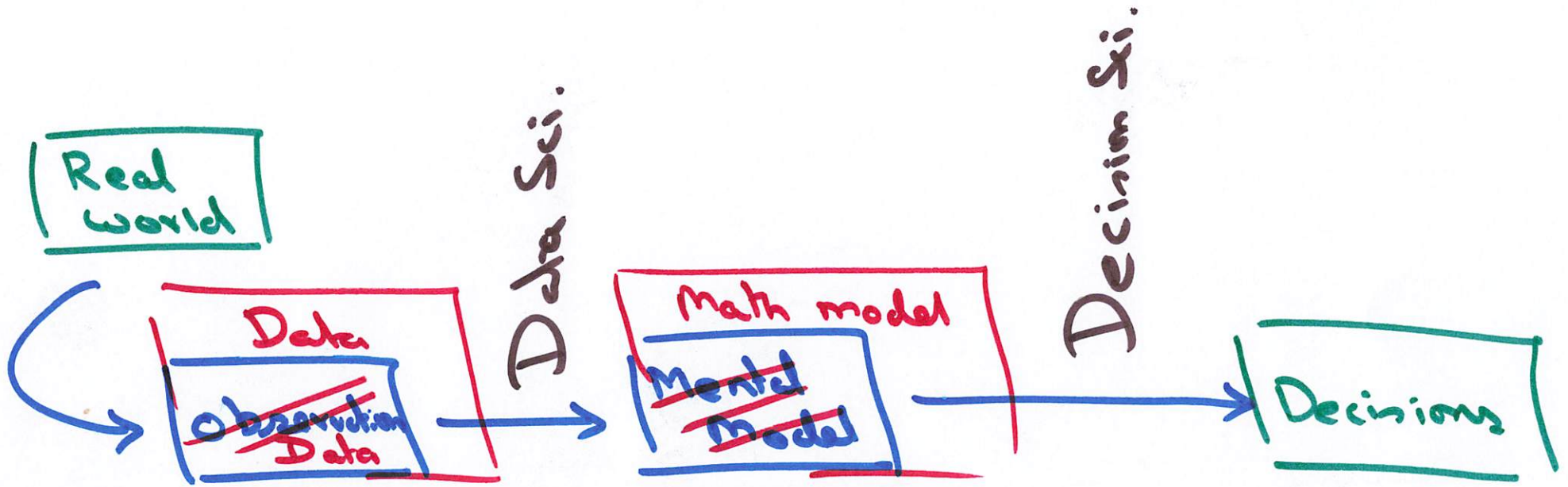
• Artificial Intelligence

• What it is?

• A brief history

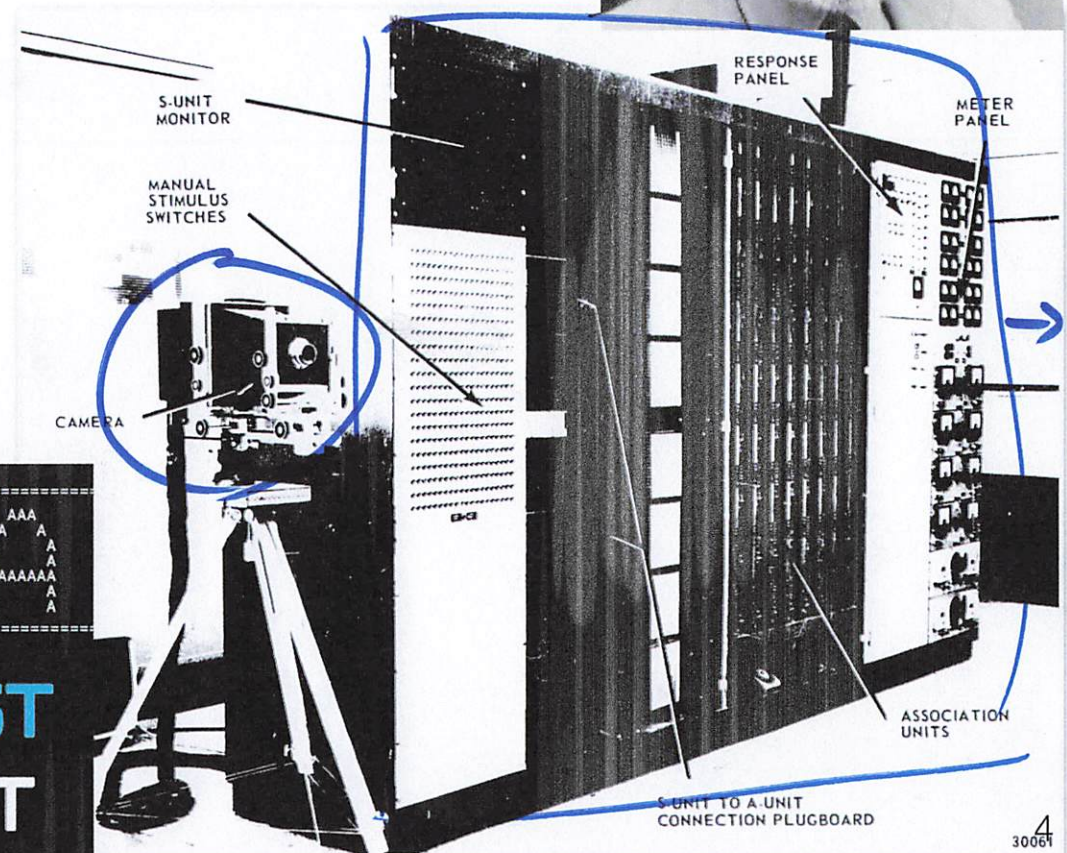
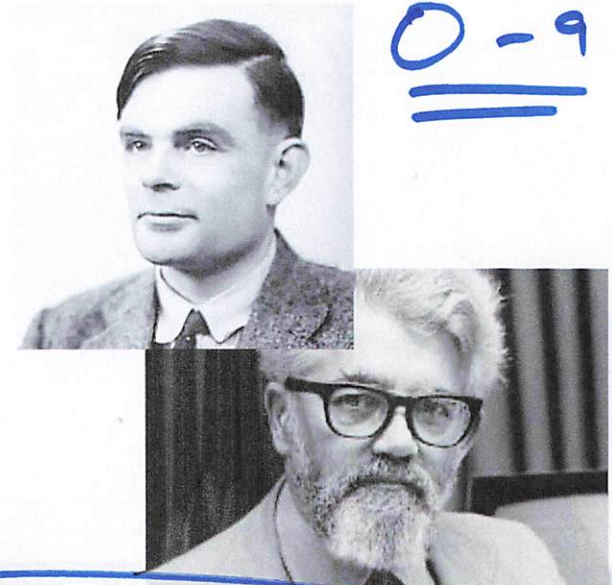
~~Built Int.~~

Fake Int.



Early days

- McCulloch-Pitts Neuron, 1940
- Alan Turing (1950): The Turing test
- John McCarthy (1956): Artificial Intelligence
- "the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence." - Rosenblatt, (1958)

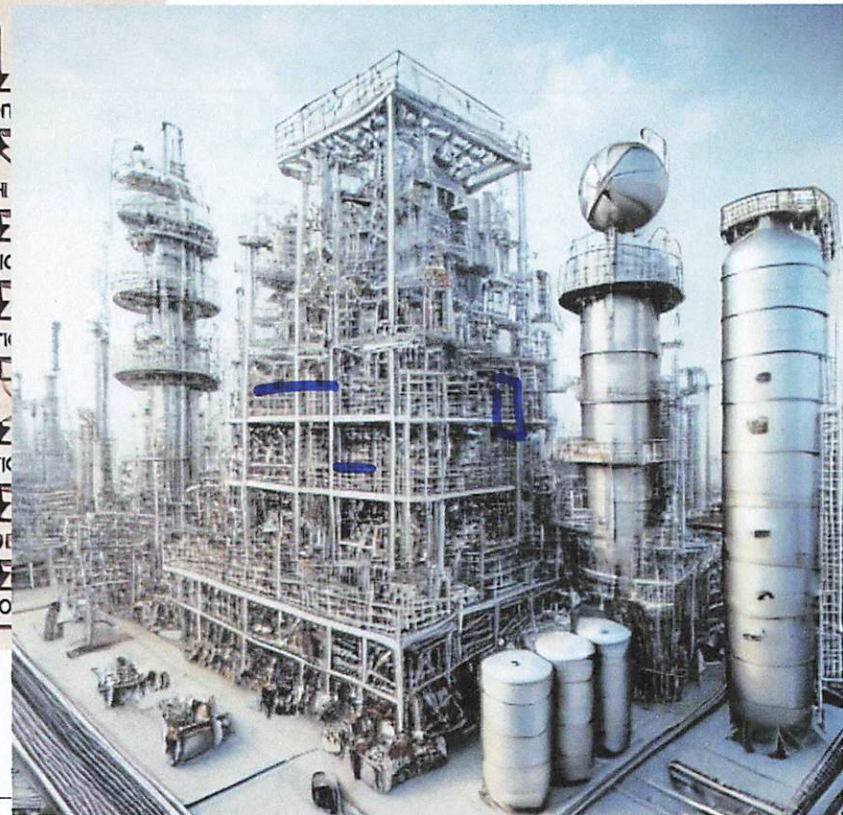
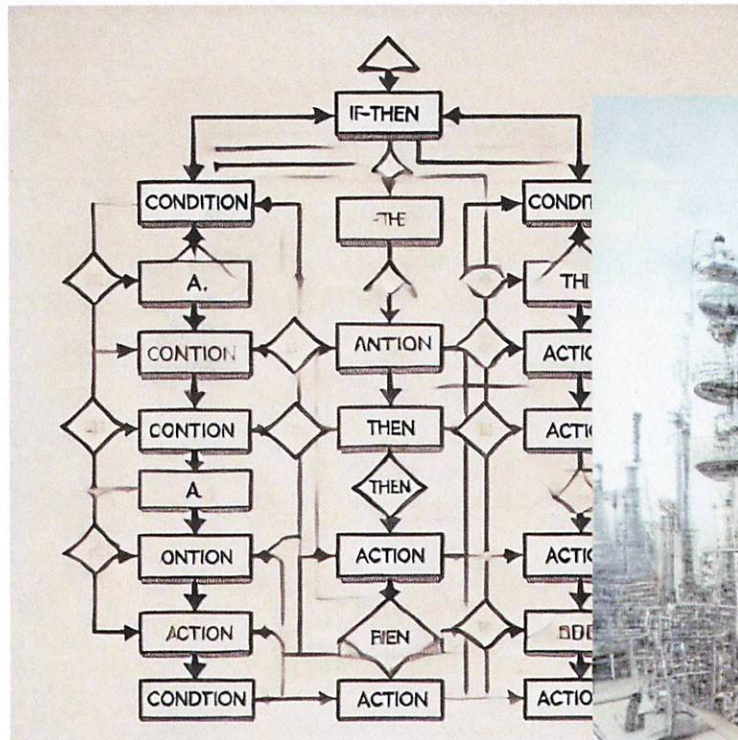


before LLMs!

1956 - 1975

1975 - 1980

- Symbolic AI (1950s - 1980s), Expert Systems (1970s - 1990s)
- Methods: Rule-based systems, knowledge representation.
- Applications: Medical diagnosis systems, financial advisory tools. Expert systems like MYCIN (diagnostic system), Dendral (chemical analysis), and SHRDLU (natural language processing).



before LLMs!

- Dynamic programming and Reinforcement Learning (1990s - abt 2020!)
 - Methods: Dynamic Programming, Stochastic Control, Markov decision processes, Q-learning, deep reinforcement learning.
 - Applications: Game playing (e.g., Deepblue, AlphaGo), recommendation systems, Autonomous robotics.

